Applications

Twelve cited real-world uses

14 July 2026

Contents

1 Part 2 — Exceptional Real-World Applications (Literature Review)

Twelve applications of population/swarm metaheuristics from peer-reviewed venues, spanning seven domains. Each entry gives (a) the problem, (b) the algorithm and why it fit, (c) the solution encoding, (d) the fitness/objective, (e) constraint handling, (f) reported results, and (g) the full citation.

Sourcing honesty. Every citation and headline number below was verified against the primary source (or, where noted, the publisher abstract/record). Figures the source did not let us confirm to the digit are tagged `[unverified]`; they are reported as the authors state them, not as our claims. This transparency is deliberate: an arXiv-bound report must not launder unverifiable numbers as fact.

1.1 Index

#	Domain	Problem	Algorithm	Year	Venue
1	Medicine / Bioinformatics	Cancer gene selection	Master–Slave Binary GWO	2021	BioMed Research International
2	Medicine	4D lung-SBRT radiotherapy planning	PSO (virtual-search)	2016	IEEE Trans. Biomed. Eng.
3	Aerospace / Structural	Spacecraft multi-gravity-assist trajectory	Self-adaptive DE	2022	Acta Astronautica

#	Domain	Problem	Algorithm	Year	Venue
4	Aerospace / Structural	Truss topology + sizing	Novelty-driven Binary PSO	2021	arXiv (Assimi et al.)
5	Energy	Distribution-net reconfiguration + DG siting	PSO	2025	PLOS ONE
6	Energy	Wind-farm micro-siting (position + hub height)	PSO	2021	Applied Sciences
7	Robotics	6-DOF welding-arm inverse kinematics	Grey Wolf Optimizer	2022	IJORAS
8	NAS	ImageNet classifier architecture search	Regularized (aging) Evolution	2019	AAAI
9	5G Wireless	Small-cell base-station siting (real field data)	Multi-objective Cuckoo Search + K-means	2023	PeerJ Comp. Sci.
10	Edge Computing	Edge-server placement	Hybrid GA + PSO	2024	PeerJ Comp. Sci.
11	Logistics	Container-terminal berth allocation	Genetic Algorithm	2018	Pesquisa Operacional
12	Logistics	Cold-chain refrigerated vehicle routing	Hybrid Tabu–GWO	2024	PLOS ONE

1.2 1 · Gene selection for cancer classification — Master–Slave Binary GWO

- **(a) Problem.** Select the few informative genes from high-dimensional microarray/biomedical datasets (thousands of features, tens of samples) so a downstream classifier diagnoses cancer accurately without noise from redundant genes.
- **(b) Algorithm + fit.** A **Master–Slave Binary Grey Wolf Optimizer (MSBGWO)**. Feature selection is a binary, high-dimensional, deceptive search (the ”small-n, large-p” regime); GWO's α/β/δ leadership gives strong exploitation, and the master–slave twist (each weak ”slave” wolf learns from an assigned ”master”) injects the diversity plain GWO lacks, escaping the local optima that trap BPSO/BGA here.
- **(c) Encoding.** Binary feature mask of length D = number of genes; bit $= 1$ keeps the gene. Continuous wolf positions are binarised through a sigmoid transfer function.
- **(d) Fitness.** $\mathrm{fit} = 1 - \alpha\frac{S}{D} - \alpha\,\overline{\mathrm{Acc}}$ with $\alpha = 0.8$, S = number of selected genes, D = total genes, $\overline{\mathrm{Acc}}$ = mean KNN (k=5) accuracy — jointly rewards accuracy and sparsity.
- **(e) Constraints.** Unconstrained wrapper; feasibility is inherent to the binary mask (any subset is valid).
- **(f) Results.** Accuracy **1.000** on Leukemia and DLBCL, **0.996** on Ovarian, **0.957** on Colon, **0.850** on CNS, while selecting the fewest features; beats BGWO2, BGA, BPSO, DE, and SCA on accuracy, precision, recall, F-measure, and feature count `[verified]`.
- **(g) Citation.** Momanyi, E.; Segera, D. ”A Master-Slave Binary Grey Wolf Optimizer for Optimal Feature Selection in Biomedical Data Classification.” *BioMed Research International*, 2021:5556941. DOI 10.1155/2021/5556941.

1.3 2 · 4D lung-SBRT radiotherapy planning — PSO with a "virtual search" strategy

- **(a) Problem.** In 4D conformal radiotherapy for lung SBRT, choose the beam aperture monitor-unit (MU) weights across all respiratory phases to deliver a lethal tumour dose while sparing healthy tissue.
- **(b) Algorithm + fit. PSO** with a novel constraint-handling ("virtual search") strategy. The dose objective is a high-dimensional, constrained, non-linear function of continuous weights with no clean gradient — PSO searches it directly; the virtual-search normalisation keeps the swarm in the clinically feasible region.
- **(c) Encoding.** A particle is the weight vector $\vartheta = [\vartheta_1, \dots, \vartheta_{N_{\text{ap}}}]$, $0 \le \vartheta_k \le 3.3$; dimension **90–110** apertures for the tested cases.
- **(d) Fitness.** Dose-based MSE $F = \sum_i^{N_{\text{struct}}} \frac{1}{N_{\text{vox},i}} \big(\sum F_{i,f}^{\text{up}} + \sum F_{i,f}^{\text{low}} + F_i^{\text{max}} \big)$ penalising over-, under-, and max-dose voxel violations; dose $\mathbf{D} = \sum_k \vartheta_k \mathbf{d}_k$.
- **(e) Constraints.** Three schemes compared; the proposed one iteratively rescales dose by $NF^t = D_{95}^{\text{presc}} / D_{95}^t$ (feasibility-restoring normalisation), vs a hard penalty of 10^{26} for violating 95 % tumour coverage at 54 Gy.
- **(f) Results.** Across 5 lung-SBRT patients the virtual-search PSO reached the smallest objective values and tightest convergence ranges; a dose-projection baseline found the global optimum in only 2/5 cases and needed ~200 vs 30 iterations `[verified]`.
- **(g) Citation.** Modiri, A.; Gu, X.; Hagan, A.M.; Sawant, A. "Radiotherapy Planning Using an Improved Search Strategy in Particle Swarm Optimization." *IEEE Trans. Biomedical Engineering*, 64(5):980–989, 2016 (landmark clinical reference; ~1 yr outside the 8-yr window). DOI 10.1109/TBME.2016.2585114.

1.4 3 · Deep-space multi-gravity-assist trajectory — Self-adaptive DE

- **(a) Problem.** Design propellant-efficient interplanetary trajectories that chain multiple planetary gravity assists plus deep-space manoeuvres — ESA's GTOP benchmark suite, built precisely because these are severely multimodal.
- **(b) Algorithm + fit. Self-adaptive ("self-learning") Differential Evolution** — mutation strategy and (F, CR) switch mid-run based on recent success, plus re-initialisation of stagnated sub-populations. DE's vector-difference mutation suits the continuous decision variables; the adaptive+restart machinery targets DE's premature-convergence weakness that makes GTOP hard.
- **(c) Encoding.** Real-valued MGA-1DSM vector: launch epoch t_0; hyperbolic excess velocity $[V_\infty, \text{RA}, \text{dec}]$; per-leg times-of-flight; per-flyby swing-by parameters; dimension $\approx 4+3(n-1)$ for n planets `[unverified – standard GTOP chromosome, not itemised in the paper text obtained]`.
- **(d) Fitness.** Minimise total mission Δv (equiv. maximise delivered mass), a highly multimodal scalar of the trajectory vector `[unverified exact form]`.
- **(e) Constraints.** Box bounds (launch window, TOF ranges, min flyby altitude); continuity satisfied by the encoding; explicit re-initialisation to escape local optima.
- **(f) Results.** Solved 6 well-known ESA GTOP problems; matched the known best on **5 of 6** and set a **new best-known on 4 of 6** `[verified headline; per-problem delta-v paywalled, unverified]`.
- **(g) Citation.** Choi, J.H.; Lee, J.; Park, C. "Deep-space trajectory optimizations using differential evolution with self-learning." *Acta Astronautica*, 191:258–269, 2022. DOI 10.1016/j.actaastro.2021.11.014.

1.5 4 · Truss topology + sizing — Novelty-driven Binary PSO

- **(a) Problem.** Minimum-weight design of trusses, choosing both topology (which members exist) and sizing (discrete cross-sections) — a combinatorial, multimodal structural-optimisation problem.
- **(b) Algorithm + fit. Novelty-driven Binary PSO** in a bilevel scheme: the upper level uses binary PSO to *discover diverse topologies by maximising novelty* (not just fitness), the lower level does discrete member sizing. The novelty drive combats the premature convergence that plagues weight-only search on this deceptive landscape.

- **(c) Encoding.** Binary particle at the upper level (member presence); discrete-catalogue section indices at the lower level.
- **(d) Fitness.** Minimise structural weight subject to stress/displacement limits; the upper level additionally maximises a novelty metric over the archive of found designs.
- **(e) Constraints.** Stress and nodal-displacement limits (standard penalty/repair in truss BPSO).
- **(f) Results.** Authors report the method "outperforms the current state-of-the-art" and returns *multiple* high-quality designs `[verified qualitative claim; the abstract obtained lists no per-truss weights]`.
- **(g) Citation.** Assimi, H.; Neumann, F.; Wagner, M.; Li, X. "Novelty-Driven Binary Particle Swarm Optimisation for Truss Optimisation Problems." arXiv:2112.07875, 2021.

1.6 5 · Distribution-network reconfiguration + DG siting — PSO (real Ethiopian feeder)

- **(a) Problem.** Jointly choose tie-switch states (reconfiguration) and the location/size of distributed generation (DG) on the **real Wolaita Sodo MV feeder, Ethiopia** (114 transformers, ≈27 MVA) to cut losses and fix severe under-voltage — not a toy IEEE 33/69-bus feeder.
- **(b) Algorithm + fit. PSO**. The search mixes a combinatorial part (switch states that must keep the feeder radial — a discontinuous graph constraint) with a continuous part (DG MW), evaluated through a backward/forward-sweep load flow that jumps discontinuously on topology change. No usable gradient exists; PSO searches switch+DG combinations directly.
- **(c) Encoding.** Each particle encodes 4 tie-switch open/closed assignments (restricted to radial/fundamental-loop combinations) plus DG bus index + MW rating; swarm $n = 20$, $I_{\text{max}} = 20$.
- **(d) Fitness.** Weighted sum $F = w_1 \Delta P_{\text{loss}} + w_2 \Delta Q_{\text{loss}} + w_3\,\text{VD}$ (real loss $\sum I^2 R$, reactive loss $\sum I^2 X$, voltage-deviation index).
- **(e) Constraints.** Radiality enforced by *restricting the search space* to spanning-tree combinations (encoding-level repair, not a penalty); voltage/ current/DG-size limits checked as hard feasibility in the load flow.
- **(f) Results.** Combined reconfig+DG cut active loss **1631.10 → 455.66 kW (−72.06 %)**, raised min voltage **0.7537 → 0.9550 p.u.**, cut max deviation **24.63 % → 4.5 %**, annual energy loss **16.669 → 4.164 GWh**, ≈16.40 M ETB/yr savings on a 22.07 M ETB investment (~6-yr payback) `[verified]`.
- **(g) Citation.** Alemayehu, B.; Mishra, S.; Tejani, G.G.; Tripathi, S. "Network reconfiguration and DG based compensation of Wolaita Sodo distribution system by using particle swarm optimisation." *PLOS ONE*, 20(10):e0335512, 2025. DOI 10.1371/journal.pone.0335512.

1.7 6 · Wind-farm micro-siting with variable hub height — PSO (real Manjil site)

- **(a) Problem.** Re-optimise the layout of Iran's **Manjil/Rudbar** wind complex by jointly choosing each turbine's (x, y) position **and** hub height, using the site's real wind regime, to cut wake losses and cost of energy.
- **(b) Algorithm + fit. PSO**. Jensen wake interactions make even the 2D layout multimodal; adding hub height as a third variable means a taller turbine can clear a shorter one's wake, so the optimum changes *discontinuously* as heights cross wake cones — a non-smooth landscape ideal for derivative-free swarm search.
- **(c) Encoding.** Per turbine a triple (x_i, y_i, h_i) — planar coordinates + hub height from a discrete menu $\{40, 50, 60\}$ m; $\approx 3 \times 21 = 63$-D for the 21-turbine case `[dimension is our derivation, unverified]`.
- **(d) Fitness.** Minimise cost of energy $CoE = (C_{\text{cap}}(h) + C_{\text{O\&M}})/AEP(x, y, h)$, with AEP from the wind-rose-weighted power curve minus pairwise Jensen wake deficits.
- **(e) Constraints.** Min inter-turbine spacing and site-boundary limits; h_i restricted to the discrete menu; infeasible layouts penalised `[penalty coefficients unverified]`.
- **(f) Results.** Optimised layout **+10.75 %** power and **−9.42 %** levelised cost vs baseline; the mixed-height 21-turbine config (2×60 m, 4×50 m, rest 40 m) produced 7.75 MW `[verified`

headline; granular AEP/$MWh omitted as unverifiable].

- **(g) Citation.** Yeghikian, M.; Ahmadi, A.; Dashti, R.; Esmaeilion, F.; Mahmoudan, A.; Hoseinzadeh, S.; Garcia, D.A. "Wind Farm Layout Optimization with Different Hub Heights in Manjil Wind Farm Using PSO." *Applied Sciences*, 11(20):9746, 2021. DOI 10.3390/app11209746.

1.8 7 · 6-DOF welding-arm inverse kinematics — Grey Wolf Optimizer

- **(a) Problem.** Solve the inverse kinematics of a new 6-DOF revolute arm for automated oil-and-gas pipeline welding: find the six joint angles driving the end-effector along a rectangular weld path — analytically messy because many joint combinations reach the same pose.
- **(b) Algorithm + fit. GWO** (vs PSO/WOA/JFO and an improved I-GWO). On the smooth-but-redundant, low-dimensional, box-constrained IK landscape, GWO's α/β/δ exploitation plus its $a : 2 \to 0$ decay converge far more reliably than the competitors.
- **(c) Encoding.** Continuous 6-D angle vector $[\theta_1, \ldots, \theta_6]$, box-bounded per Denavit–Hartenberg limits; population 50, 10 iterations, solved per waypoint over 6 waypoints.
- **(d) Fitness.** Minimise position MSE $= \frac{1}{N} \sum_{i \in \{x,y,z\}} (P_i^{\text{ref}} - P_i)^2$ (forward-kinematics vs reference).
- **(e) Constraints.** Hard box-clipping of each joint angle to its DH range during updates.
- **(f) Results.** GWO per-waypoint MSE **8.4×10^{-7}–6.5×10^{-5}**; I-GWO best (**1.6×10^{-7}–9.8×10^{-7}**); PSO 2–4 orders worse (**1.7×10^{-3}–3.4×10^{-3}**); only GWO/I-GWO "effectively solved" the task [verified].
- **(g) Citation.** Nyong-Bassey, B.E.; Epemu, A.M. "Inverse Kinematics Analysis of Novel 6-DOF Robotic Arm Manipulator for Oil and Gas Welding Using Meta-Heuristic Algorithms." *IJORAS*, 4:13–22, 2022. DOI 10.33093/ijoras.2022.4.3.

1.9 8 · ImageNet architecture search — Regularized (aging) Evolution → AmoebaNet

- **(a) Problem.** Automatically discover a convolutional image classifier that beats the best human- and reinforcement-learning-designed architectures on ImageNet under a fixed search budget.
- **(b) Algorithm + fit. Regularized ("aging") Evolution** — a tournament-selection GA that each cycle discards the *oldest* member (not the worst), biasing survival toward genotypes with sustained recent success. Suits the huge, discrete, non-differentiable architecture space and, at equal compute, beat RL controllers — a genuinely non-obvious result.
- **(c) Encoding.** Two cell types (normal, reduction), each a sequence of 5 blocks; each block picks 2 hidden states + one op each from a discrete op set; network = stacked cells. Genome = discrete (state, state, op, op) tuples.
- **(d) Fitness.** Validation top-1 accuracy after a truncated training proxy; population $P = 100$, tournament sample $S = 25$, aging removal.
- **(e) Constraints.** No penalty — structural validity guaranteed by the mutation operator (one pointer/op changed at a time); age-based culling is the regulariser.
- **(f) Results.** AmoebaNet-A **82.8 %** top-1 **/ 96.1 %** top-5 at 86.7 M params (≈ NASNet-A RL 82.7/96.2 at 88.9 M); scaled to 469 M → **83.9 % / 96.6 %** ImageNet SOTA [verified]; the "evolution beats RL faster early" claim [unverified — no exact multiplier].
- **(g) Citation.** Real, E.; Aggarwal, A.; Huang, Y.; Le, Q.V. "Regularized Evolution for Image Classifier Architecture Search." *AAAI-19*, 2019. arXiv:1802.01548; DOI 10.1609/aaai.v33i01.33014780.

1.10 9 · 5G small-cell base-station siting — Multi-objective Cuckoo Search + K-means

- **(a) Problem.** Site 5G small cells across real neighbourhoods of **Belém, Brazil** to maximise coverage while minimising transmit power, driven by real OpenCellID field data rather than a synthetic grid.
- **(b) Algorithm + fit. MOCS-KM** (multi-objective Cuckoo Search + K-means). Cuckoo Search's heavy-tailed Lévy flights escape the many local optima of a noisy real-field cover-

age surface; K-means turns the metaheuristic's choices into concrete cell clusters/powers. (Closest published cousin of this lab's own Wi-Fi placement project.)

- **(c) Encoding.** Low-dimensional mixed vector $[k, P_t]$: k = number of cells (integer 1–100) seeding K-means over real coordinates; P_t = transmit power (continuous, 30–40 dBm).
- **(d) Fitness.** $f = 0.7\, N_{\text{outage}} + 0.3\, P_{\text{diff}}$ (percent users in outage vs power above the 30 dBm floor).
- **(e) Constraints.** $P_r \geq -90$ dBm defines connected-vs-outage, enforced in fitness; k, P_t box-bounded.
- **(f) Results.** **0 %** outage @700 MHz (12 cells, 30 dBm), **2.43 %** @2.3 GHz, **2.29 %** @3.5 GHz; max coverage **97.4 %**; runtime 552–993 s `[verified]`.
- **(g) Citation.** Ferreira, F.H.; Barros, F.J.B.; de Alcântara Neto, M.C.; Cardoso, E.; Francês, C.R.L.; Araújo, J. "Hybrid computational and real data-based positioning of small cells in 5G networks." *PeerJ Computer Science*, 2023. DOI 10.7717/peerj-cs.1412.

1.11 10 · Edge-server placement — Hybrid GA + PSO (GP4ESP)

- **(a) Problem.** Decide which base/edge stations to equip with limited edge servers so that overall request **response time** is minimised in a heterogeneous edge-computing network — crucially accounting for server *computing delay*, which prior work ignored (over-estimating service quality).
- **(b) Algorithm + fit.** A **hybrid GA + PSO** that fuses PSO's swarm cognition into GA's evolutionary operators. ESP is a combinatorial assignment problem with a rugged objective; the hybrid balances GA's global recombination with PSO's directed convergence.
- **(c) Encoding.** Selection/assignment representation over candidate edge sites (which stations host servers) `[standard ESP encoding; the paper's exact vector layout not quoted here]`.
- **(d) Fitness.** Minimise overall response time = network transmission delay + edge-server computing/queueing delay (the paper's key modelling addition).
- **(e) Constraints.** Limited edge-resource budget (number/capacity of servers).
- **(f) Results.** **18.2 %–20.7 %** shorter overall response time vs **eleven** up-to-date ESP algorithms, stable as problem scale varies `[verified via publisher record/PMC]`.
- **(g) Citation.** Han, et al. "GP4ESP: a hybrid genetic algorithm and particle swarm optimization algorithm for edge server placement." *PeerJ Computer Science*, 2024. DOI 10.7717/peerj-cs.2439 (PMC11623000).

1.12 11 · Container-terminal berth allocation — Genetic Algorithm

- **(a) Problem.** Discrete berth allocation at the real **TECON Rio Grande** container terminal (Brazil): assign arriving vessels to berths and fix berthing order/time to minimise service delay and time-window penalties.
- **(b) Algorithm + fit.** **GA** (heuristic crossover), benchmarked head-to-head vs Simulated Annealing. GA is a population-based global search over the combinatorial, multimodal berthing-sequence space; SA is the single-point foil.
- **(c) Encoding.** Permutation chromosome — the ordered vessel sequence per berth; primary case 62 vessels over 3 berths (also 35/55/62 vessels × 5/10 berths).
- **(d) Fitness.** Minimise $Z^* = w_0 \sum v_i(\cdot) + w_1 \sum [\max(0, a_i - T) + \max(0, T + t - b_i)] + w_2 \sum [\cdot]$ (service cost + vessel and berth time-window penalties), weights $[1, 10, 10]$.
- **(e) Constraints.** Penalty relaxation — hard time-window constraints become soft penalty terms, allowing temporarily infeasible intermediate solutions.
- **(f) Results.** Primary case (30 runs): GA avg **185,951** vs SA **194,074** (**~4.2 %** lower), GA **20′50″** vs SA **29′04″** (**~28 %** faster), GA std **2,202** vs SA **3,267**; GA ≥ SA across all six scenarios `[verified]`.
- **(g) Citation.** Pereira, E.D.; Coelho, A.S.; Longaray, A.A.; Machado, C.M.S.; Munhoz, P.R. "Metaheuristic Analysis Applied to the Berth Allocation Problem: Case Study in a Port Container Terminal." *Pesquisa Operacional*, 38(2), 2018. DOI 10.1590/0101-7438.2018.038.02.0247.

1.13 12 · Cold-chain refrigerated vehicle routing — Hybrid Tabu–GWO

- **(a) Problem.** Route refrigerated vehicles for a real fresh-food e-commerce operator in China under vehicle-capacity, driving-range, dual-time-window, and perishable-freshness constraints.
- **(b) Algorithm + fit. Hybrid Tabu Search – Grey Wolf Optimizer (TGWO)**. GWO converges fast but has weak local search; grafting Tabu neighbourhood moves (insert/swap/2-opt) sharpens exploitation and escapes local optima on the multi-cost cold-chain landscape.
- **(c) Encoding.** Each wolf is a TSP-style permutation from the depot, decoded into multi-vehicle routes by sequential assignment; case dimension $D = 19$ (1 depot + 18 customers).
- **(d) Fitness.** Minimise $Z = C_1 + C_2 + C_3 + C_4$ (fixed vehicle + transport incl. refrigeration energy + cargo spoilage + time-window penalty).
- **(e) Constraints.** Large-coefficient penalty ($M{=}10^8$) for time-window breach; a repair-like decomposition opens a new route when capacity/range/ freshness would be violated.
- **(f) Results.** Total cost TS **17,314.8**, GWO **7,754.9**, TGWO **7,112.5**; TGWO saves **50.34 %** / **30.66 %** in travel distance vs TS / GWO `[verified; the authors' "143.44 %" cost-saving figure uses a non-standard denominator and is flagged, not endorsed]`.
- **(g) Citation.** Zhang, H.; Yan, J.; Wang, L. ”Hybrid Tabu-Grey wolf optimizer algorithm for enhancing fresh cold-chain logistics distribution.” *PLOS ONE*, 19(8):e0306166, 2024. DOI 10.1371/journal.pone.0306166.

1.14 Ranking — the top 3 most exceptional

#1 — AmoebaNet / Regularized Evolution (App 8). The clever trick is *age-based regularization*: culling the oldest genotype instead of the worst turns a plain GA into a search that resists overfitting to lucky early winners. It was the first evolved architecture to *beat* both human- and RL-designed networks at ImageNet scale (83.9 % top-1), overturning the assumption that reinforcement learning was the right NAS paradigm. The mechanism is one line of code, and it changed how AutoML search is done.

#2 — Wolaita Sodo reconfiguration + DG (App 5). The clever move is *constraint-as-encoding*: rather than penalising non-radial grids, the search space itself is restricted to spanning-tree/fundamental-loop combinations, so every particle is feasible by construction — eliminating an entire class of wasted evaluations. Coupled with a real deployed 114-transformer Ethiopian feeder and audited economics (72 % loss cut, ~6-year payback in real ETB), it is the entry with the hardest real-world grounding and the largest tangible impact.

#3 — Manjil variable-hub-height wind siting (App 6). The clever trick is the *encoding*: promoting hub height to a free decision variable per turbine lets a tall turbine physically climb out of an upstream turbine's wake — turning a 2-D layout problem into a 3-D one where the extra dimension buys energy that no same-height layout can. It is a rare case where enlarging the search space *reduces* effective difficulty, and it is validated on a real, historic wind farm (+10.75 % power, −9.42 % cost).